

DS	Session	Linked List	Question Information	Level 1	Challenge 31
Question description Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit. He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test. Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked. For example if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25. Constraints 1 < N < 100 1 < D < 1000 Input Format First line contains the number of datas- N. Second line contains N integers(the given linked list). Next line indicates the node data D that has to be deleted. Output Format Single line represents the linked list after required elements deleted.					
Logical Test Cases Test Case 1 INPUT (STDIN) 6 7 9 8 8 4 2 8 EXPECTED OUTPUT Linked List:->7->9->4->2 Test Case 2 INPUT (STDIN) 10 5 3 7 2 7 9 8 8 4 2 2 EXPECTED OUTPUT Linked List:->5->3->7->7->9->8->8->4					

▼ Mandatory Test Cases

Test Case 1

KEYWORD

`struct node`

Test Case 2

KEYWORD

`node *next;`

Test Case 3

KEYWORD

`void create()`

Test Case 4

KEYWORD

`p2=p2->next;`

Test Case 5

KEYWORD

`void del()`

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

280

Test Case 3

NLOC

65

```
LinkedList1.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *start = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node*)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (start == NULL) {
16         start = temp;
17     } else {
18         struct node *p = start;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```
26 void del(int key) {
27     while (start != NULL && start->data == key) {
28         struct node *temp = start;
29         start = start->next;
30         free(temp);
31     }
32     struct node *p2 = start;
33     while (p2 != NULL && p2->next != NULL) {
34         if (p2->next->data == key) {
```

main | CH.SC.U4CSE25225 | LinkedList1.c

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```
36         p2->next = temp->next;
37         free(temp);
38     } else {
39         p2 = p2->next;
40     }
```

```
41     }
42 }
43
44 void display() {
45     printf("Linked List:");
46     struct node *p = start;
47     while (p != NULL) {
48         printf("->%d", p->data);
49         p = p->next;
50     }
51     printf("\n");
52 }
53
54 int main() {
```

```
55  int n;
56  if (scanf("%d", &n) != 1) return 0;
57  for (int i = 0; i < n; i++) {
58      int val;
59      scanf("%d", &val);
60      create(val);
61  }
62  int key;
63  scanf("%d", &key);
64  del(key);
65  display();
66  return 0;
67}
68
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
7 9 8 8 4 2
```

```
8
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList1.exe
```

```
Linked List:->7->9->4->2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
10
```

```
5 3 7 2 7 9 8 8 4 2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList1.exe
```

```
Linked List:->5->3->7->7->9->8->8->4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Linked List	Question Information	Level 1	Challenge 32
Question description saran, subash, and Yasir alias Pari are three first-year engineering students of the State Technical Institution (STI), India. While saran and subash are average students who come from a Middle class, Yasir is from a rich family. saran studies, engineering as per his father's wishes, while subash, whose family is poor, studies engineering to improve his family's financial situation. Yasir, however, studies engineering of his simple passion for developing android applications. Yasir is participating in a hackathon for android application development. the task is Insertion in a Doubly Linked list at beginig. Functional Description: In the doubly linked list, we would use the following steps to insert a new node at the beginning of the doubly linked list. • Create a new node • Assign its data value • Assign newly created node's next ptr to current head reference. So, it points to the previous start node of the linked list address • Change the head reference to the new node's address. • Change the next node's previous pointer to new node's address (head reference) Constraints $0 < N < 100$ $0 < arr < 1000$					

Input Format

First line indicates the number of elements N to be inserted in array

Second line indicates the array elements according to the N

Output Format

First line represents the doubly linked list in forward direction

Second Line represents the doubly linked list in backward direction

Logical Test Cases

Test Case 1

INPUT (STDIN)

16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

EXPECTED OUTPUT

17 98 32 56 24 23 19 18 14 13 11 9 7 4 3 1
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

Test Case 2

INPUT (STDIN)

15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

EXPECTED OUTPUT

2 1 0 2 2 0 2 0 1 1 1 0 2 1 0
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

Mandatory Test Cases

Test Case 1

KEYWORD

struct Node

Test Case 2

KEYWORD

struct Node *next;

Test Case 3

KEYWORD

struct Node *prev;

Test Case 4

KEYWORD

```
void insertStart(struct  
Node** head,int data)
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

269

Test Case 3

NLOC

47

LinkedList2.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7     struct Node *prev;
8 };
9
10 void insertStart(struct Node** head, int data) {
11     struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
12     new_node->data = data;
13     new_node->next = (*head);
14     new_node->prev = NULL;
15     if ((*head) != NULL) {
16         (*head)->prev = new_node;
17     }
18     (*head) = new_node;
19 }
20
21 int main() {
22     int n;
23     if (scanf("%d", &n) != 1) return 0;
24
25     struct Node* head = NULL;
```

```
26     for (int i = 0; i < n; i++) {
27         int val;
28         scanf("%d", &val);
29         insertStart(&head, val);
30     }
31
32     struct Node* curr = head;
33     struct Node* tail = NULL;
34     while (curr != NULL) {
35         printf("%d ", curr->data);
36         if (curr->next == NULL) tail = curr;
37         curr = curr->next;
```

```
38     }
39     printf("\n");
40
41     curr = tail;
42     while (curr != NULL) {
43         printf("%d ", curr->data);
44         curr = curr->prev;
45     }
46     printf("\n");
47     return 0;
48 }
49
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

16

1 3 4 7 9 11 13 14 18 19 23 24

17 98 32 56 24 23 19 18 14 13

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList2.exe

56 32 98 17 24 23 19 18 14 13 11 9 7 4 3 1

1 3 4 7 9 11 13 14 18 19 23 24 17 98 32 56

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

15

56 32 98 17 0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList2.exe

2 0 2 0 1 1 1 0 2 1 0 17 98 32 56

56 32 98 17 0 1 2 0 1 1 1 0 2 0 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

Question description

Varman's Dream came true after he got an Appointment order from Google. Simon's family was very happy of his achievement.

The company mentioned Basic Salary, DA, HRA with some other benefits.

But not highlighted the Gross salary in the order.

varman's father wanted to know the Gross salary of his son.

varman try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, Sorted insert in circular linked list.

Can you help varman?

Function Description

First case one is if linked list is empty then since new_node is only node in circular linked list, make a self loop.and change the head pointer to the new_node pointer.

Second case is new node insert in starting or before the head node.

A- Find out the last node using a loop .

```
While(present->!=*head_ref)
```

```
present=present->next;
```

B- Change the next of last node;

```
present->next=new-node;
```

C- Change next of new node to point to head.

```
new_node->next=*head_ref;
```

D- Change the head pointer to point to new node.

```
*head_ref=new_node;
```

Third case is when we insert the new node after the head in any position,then

A- Locate the node after which new node is to be inserted.

```
while(present->next!= *head_ref &&
present->next->data data)
{ present = present->next; }
```

B- Make next of new_node as next of the located pointer

```
new_node->next = present->next;
```

C- Change the next of the located pointer

```
present->next = new_node;
```

Constraints

$0 < n < 100$

Input Format:

The First line of the input represents the number of elements

Second line represents the elements of circular linked list

Output Format:

single line prints the results as per sample test cases

Logical Test Cases

Test Case 1

INPUT (STDIN)

6
10 20 30 45 34 56

EXPECTED OUTPUT

10 20 30 34 45 56

Test Case 2

INPUT (STDIN)

6
1 2 11 12 56 90

EXPECTED OUTPUT

1 2 11 12 56 90

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
struct Node *next;
```

Test Case 2

KEYWORD

```
void sortedInsert(struct  
Node** head_ref, struct  
Node* new_node)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

6

Test Case 2

TOKEN COUNT

345

Test Case 3

NLOC

68

LinkedList3.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node {
5     int data;
6     struct Node *next;
7 };
8
9 void sortedInsert(struct Node** head_ref, struct Node* new_node) {
10     struct Node* current;
11     if (*head_ref == NULL || (*head_ref)->data >= new_node->data) {
12         new_node->next = *head_ref;
13         *head_ref = new_node;
14     } else {
15         current = *head_ref;
16         while (current->next != NULL && current->next->data < new_node->data) {
17             current = current->next;
18         }
19         new_node->next = current->next;
20         current->next = new_node;
21     }
22 }
23
24 int main() {
25     int n;
```

main | CH.SC.U4CSE25225 | LinkedList3.c

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```
26     if (scanf("%d", &n) != 1) return 0;
27
28     struct Node* head = NULL;
29     for (int i = 0; i < n; i++) {
30         int val;
31         scanf("%d", &val);
32         struct Node* new_node = (struct Node*)malloc(sizeof(struct Node));
33         new_node->data = val;
34         new_node->next = NULL;
35         sortedInsert(&head, new_node);
```

```
36     }
37
38     struct Node* temp = head;
39     while (temp != NULL) {
40         printf("%d%c", temp->data, (temp->next == NULL) ? '\n' : ' ');
41         temp = temp->next;
42     }
43     return 0;
44 }
45
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

6

10 20 30 45 34 56

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList3.exe

10 20 30 34 45 56

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

6

1 2 11 12 56 90

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList3.exe

1 2 11 12 56 90

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Linked List	Question Information	Level 1	Challenge 34
Question description					
Problem Description:					
For Engineering Second -year students, the instructor is organising a surprise test.					
He's given you a new node is added before the given node of the given Linked List.					
For example if the given Linked List is 5->10->15->20->25 and					
we add an item 30 before node 15, then the Linked List becomes 5->10->30->15->20->25.					
Since a Linked List is typically represented by the head of it, we have to traverse the list till node and then insert the node.					
Constraints:					
1 < arr <100					
Input Format					
First line contains the number of datas- N.					
Second line contains N integers(the given linked list).					
Third line contains the node P before which the node to be inserted.					
Fourth line contain the node X to be inserted.					
Output Format					
case 1 (node P found in Linked list) :					

Display the final Linked List.

case 2(node P not found in Linked list) :

Print Node not found!

Display the Linked list.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
115 210 116 110
116
17

EXPECTED OUTPUT

Linked List:->115->210->17->116->110

Test Case 2

INPUT (STDIN)

4
15 20 16 10
25
17

EXPECTED OUTPUT

Node not found!
Linked List:->15->20->16->10

▼ Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

node *next;

Test Case 3

KEYWORD

*start

Test Case 4

KEYWORD

p2->next=p1;

Test Case 5

KEYWORD

void display()

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

275

Test Case 3

NLOC

66

LinkedList4.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *start = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node*)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (start == NULL) {
16         start = temp;
17     } else {
18         struct node *p = start;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```
26 void insertBefore(int target, int newVal) {
27     if (start == NULL) {
28         printf("Node not found!\n");
29         return;
30     }
31
32     struct node *p1 = (struct node*)malloc(sizeof(struct node));
33     p1->data = newVal;
34     p1->next = NULL;
35
36     if (start->data == target) {
37         p1->next = start;
38         start = p1;
39     }
    return;
}
```

```
40 }
```

```
42 struct node *p2 = start;
43 while (p2->next != NULL && p2->next->data != target) {
44     p2 = p2->next;
45 }
46
47 if (p2->next == NULL) {
48     printf("Node not found!\n");
49     free(p1);
50 } else {
51     p1->next = p2->next;
52     p2->next = p1;
53 }
```

```
54}  
55  
56void display() {  
57    printf("Linked List:");  
58    struct node *p = start;  
59    while (p != NULL) {  
60        printf("->%d", p->data);  
61        p = p->next;  
62    }  
63    printf("\n");  
64}  
65  
66int main() {  
67    int n;
```

```
68     if (scanf("%d", &n) != 1) return 0;
69     for (int i = 0; i < n; i++) {
70         int val;
71         scanf("%d", &val);
72         create(val);
73     }
74     int target, newVal;
75     scanf("%d %d", &target, &newVal);
76     insertBefore(target, newVal);
77     display();
78     return 0;
79 }
80
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

4

115 210 116 110

116

17

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList4.exe

Linked List:->115->210->17->116->110

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

4

15 20 16 10

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList4.exe

Node not found!

Linked List:->15->20->16->10

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Linked List	Question Information	Level 1	Challenge 35
Question description Arulmozhivarman's Dream came true after he got an Appointment order from Google.Simon's family was very happy of his achievement. The company mentioned Basic Salary, DA, HRA with some other benefits. But not highlighted the Gross salary in the order. Arulmozhivarman's father wanted to know the Gross salary of his son. Arulmozhivarman try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, to delete alternate nodes of a Linked List. In this program we want to remove alternate nodes from the singly linked list which is start from the second node of the linked list and remove all the alternate nodes of it. Can you help Arulmozhivarman? Function Description • First Take two pointers a and b. • Let initially a points to head and b points to pointer of a ,that is second node . • Than make the link of a to point to pointer of b and free b. • Next is move to a to its pointer that is next node. • Next is move b to next node of a. • Continue loop until a and b becomes NULL. Constraints $0 < n < 100$					

Input Format:
The First line of the input represents the linked list elements count

Output Format:
single line prints the elements after the deletion of alternate nodes of a Linked List

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

10

EXPECTED OUTPUT

1 3 5 7 9

Test Case 2

INPUT (STDIN)

20

EXPECTED OUTPUT

1 3 5 7 9 11 13 15 17 19

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void insert_Data(struct node
**head)

Test Case 2

KEYWORD

void delete_Alt(struct node
**head)

▼ Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

3

TOKEN COUNT

300

NLOC

68

```
LinkedList5.c +
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 void insert_Data(struct node **head, int data) {
10     struct node *newNode = (struct node*)malloc(sizeof(struct node));
11     newNode->data = data;
12     newNode->next = NULL;
13     if (*head == NULL) {
14         *head = newNode;
15         return;
16     }
17     struct node *temp = *head;
18     while (temp->next != NULL) {
19         temp = temp->next;
20     }
21     temp->next = newNode;
22 }
23
24 void delete_Alt(struct node **head) {
25     if (*head == NULL) return;
```

```
26 struct node *prev = *head;
27 struct node *node = (*head)->next;
28
29 while (prev != NULL && node != NULL) {
30     prev->next = node->next;
```

main | CH.SC.U4CSE25225 | LinkedList5.c

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```
32     prev = prev->next;
33     if (prev != NULL) {
34         node = prev->next;
35     }
36 }
37 }
```

```
38
39 void printList(struct node *head) {
40     struct node *temp = head;
41     while (temp != NULL) {
42         printf("%d%c", temp->data, (temp->next == NULL) ? '\n' : ' ');
43         temp = temp->next;
44     }
45 }
46
47 int main() {
48     int n;
49     if (scanf("%d", &n) != 1) return 0;
```

```
50
51  struct node *head = NULL;
52  for (int i = 1; i <= n; i++) {
53      insert_Data(&head, i);
54  }
55
56  delete_Alt(&head);
57  printList(head);
58  return 0;
59}
60
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
10
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList5.exe
```

```
1 3 5 7 9
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
20
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList5.exe
```

```
1 3 5 7 9 11 13 15 17 19
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

DS	Session	Linked List	Question Information	Level 1	Challenge 36
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Question description

A long time ago, there was a desolate village in India. The ancient buildings, streets, and businesses were deserted. The windows were open, and the stairwell was in disarray. You can be sure that it will be a fantastic area for mice to romp about in! People in the community have now chosen to provide high-quality education to young people in order to further the village's growth.

As a result, they established a programming language coaching centre. People from the coaching centre are presently performing a test. Create a programme for the GetNth() function, which accepts a linked list and an integer index and returns the data value contained in the node at that index position.

Example

Input: 1->10->30->14, index = 2

Output: 30

The node at index 2 is 30

Constraints

1 < N < 1000

1 < X < 1000

1 < I < 1000

Input Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third Line Index I

Output Format

First Line indicates the linked list

second line indicates the node at indexing position

Logical Test Cases

Test Case 1

INPUT (STDIN)

4
1 2 3 4
2

EXPECTED OUTPUT

Linked list:-->4-->3-->2-->1
Node at index=2:3

Test Case 2

INPUT (STDIN)

4
11 21 31 41
3

EXPECTED OUTPUT

Linked list:-->41-->31-->21-->11
Node at index=3:21

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

struct node *next;

Test Case 3

KEYWORD

int GetNth(struct node* head,int index)

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

10

TOKEN COUNT

385

NLOC

74

LinkedList6.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node* createNode(int data) {
10     struct node* new_node = (struct node*)malloc(sizeof(struct node));
11     new_node->data = data;
12     new_node->next = NULL;
13     return new_node;
14 }
15
16 void insertEnd(struct node** head, int data) {
17     struct node* new_node = createNode(data);
18     if (*head == NULL) {
19         *head = new_node;
20         return;
21     }
22     struct node* temp = *head;
23     while (temp->next != NULL) {
24         temp = temp->next;
25     }
```

```
26     temp->next = new_node;
27 }
28
29 int GetNth(struct node* head, int index) {
30     struct node* current = head;
31     int count = 0;
32     while (current != NULL) {
33         if (count == index) {
34             return current->data;
35         }
36         count++;
37         current = current->next;
38     }
39     return -1;
40 }
```

41

42 void reverseList(struct node** head) {

main | CH.SC.U4CSE25225 | LinkedList6.c

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43 struct node* prev = NULL;

44 struct node* current = *head;

45 struct node* next = NULL;

46 while (current != NULL) {

47 next = current->next;

48 current->next = prev;

49 prev = current;

50 current = next;

51 }

52 *head = prev;

53 }

54

55 void printList(struct node* head) {

```
56     struct node* p = head;
57     while (p != NULL) {
58         printf("%d%c", p->data, (p->next == NULL) ? '\n' : ' ');
59         p = p->next;
60     }
61 }
62
63 int main() {
64     int n;
65     if (scanf("%d", &n) != 1) return 0;
66
67     struct node* head = NULL;
68     for (int i = 0; i < n; i++) {
69         int val;
```

```
70     scanf("%d", &val);
71     insertEnd(&head, val);
72 }
73
74 int index;
75 scanf("%d", &index);
76 printf("Node at index=%d:%d\n", index, GetNth(head, index));
77
78 reverseList(&head);
79 printf("Reversed:");
80 printList(head);
81 return 0;
82}
83
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

4

1 2 3 4

2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList6.exe

Node at index=2:3

Reversed:4 3 2 1

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

4

11 21 31 41

3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList6.exe

Node at index=3:41

Reversed:41 31 21 11

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

DS	Session	Linked List	Question Information	Level 1	Challenge 37
----	---------	-------------	----------------------	---------	--------------

Question description

the popular engineering college got lowest pass percentage in last semester. the principal conducted faculty meeting and decided to visit all the classes surprisingly.

Dr.Ramprasath is a faculty, who handling data structure course for EEE department second year students.

one day this faculty was handling very interesting topic in data structure such that Linked List,

During this lecture time, principal surprisingly visited to the class and asking to conduct surprise test on Linked list concept.

So the faculty decided to conduct test on the topic of Linked List.

the question was given to last bench students that is,

The nodes are deleted before a certain given node in the linked list.

For example if the given Linked List is 5->10->15->20->25 and

delete before 15 then the Linked List becomes 15->20->25.

Constraint :

$1 < N < 1000$

$1 < P < N-1$

INPUT Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third line contains position of the node to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

```
4
9 77 12 6
12
```

EXPECTED OUTPUT

Linked List:->12->6

Test Case 2

INPUT (STDIN)

```
4
91 77 12 61
2
```

EXPECTED OUTPUT

Invalid Node! Linked List:->91->77->12->61

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 2

KEYWORD

```
node *next;
```

Test Case 3

KEYWORD

```
void create()
```

Test Case 4

KEYWORD

```
for(i=0;i<n;i++)
```

Test Case 5

KEYWORD

```
p1=p1->next
```

Test Case 6

KEYWORD

```
void del()
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC

58

LinkedList7.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *start = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node*)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (start == NULL) {
16         start = temp;
17     } else {
18         struct node *p = start;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```
26 void del_before(int key) {
27     if (start == NULL || start->next == NULL) {
28         printf("Invalid Node!\n");
29         return;
30     }
31
32     if (start->next->data == key) {
33         struct node *temp = start;
34         start = start->next;
35         free(temp);
36         return;
37     }
38
39     struct node *p1 = start;
```

```
40 struct node *p2 = start->next;
```

main | CH.SC.U4CSE25225 | LinkedList7.c

Ln 80, Col 1 Spaces: 4 UTF-8 CRLF C

```
42 while (p2 != NULL && p2->data != key) {  
43     p1 = p1->next;  
44     p2 = p2->next;  
45 }  
46  
47 if (p2 == NULL) {  
48     printf("Invalid Node!\n");  
49     return;  
50 }  
51  
52 p1->next = p2;  
53 free(p2);
```

```
54}  
55  
56void display() {  
57    printf("Linked List:");  
58    struct node *p = start;  
59    while (p != NULL) {  
60        printf("->%d", p->data);  
61        p = p->next;  
62    }  
63    printf("\n");  
64}  
65  
66int main() {  
67    int n;
```

```
68     if (scanf("%d", &n) != 1) return 0;
69     for (int i = 0; i < n; i++) {
70         int val;
71         scanf("%d", &val);
72         create(val);
73     }
74     int key_pos;
75     scanf("%d", &key_pos);
76     del_before(key_pos);
77     display();
78     return 0;
79 }
80
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
4
```

```
9 77 12 6
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList7.exe
```

```
Linked List:->9->77->12->6
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
4
```

```
91 77 12 61
```

```
2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList7.exe
```

```
Invalid Node!
```

```
Linked List:->91->77->12->61
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```


DS	Session	Linked List	Question Information	Level 1	Challenge 38
----	---------	-------------	----------------------	---------	--------------

Question description

Malar, subash, and Yasir alias Pari are three first-year engineering students of the State Technical Institution (STI), India. While Malar and subash are average students who come from a Middle class, Yasir is from a rich family. Malar studies, engineering as per his father's wishes, while subash, whose family is poor, studies engineering to improve his family's financial situation.

Yasir, however, studies engineering of his simple passion for developing android applications.

Yasir is participating in a hackathon for android application development. the task is foling the linked list and also need to follow the given procedure.

- Firstly, Initialize a linked list.
- Then find the middle point using two pointers at different speeds through the sequence of values until they both have equal values.
- Divide the linked list into two halves using these pointer named as slow and fast, then find middle.
- Now reverse the second half of the linked list.
- At last alternate merge of first and second halves.
- Print the fold linked list.

Constraints

$0 < N < 100$

$0 < arr < 1000$

Input Format

First line indicates the number of elements N to be inserted in array

Second line indicates the array elements according to the N

Output Format

First line represents the d linked list in original order

Second Line represents the linked list in after folding

Logical Test Cases

Test Case 1

INPUT (STDIN)

16
1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17

EXPECTED OUTPUT

Link list data:1 3 4 7 9 11 13 14 18 19 23 24 56 32 98 17
Link list data after fold:1 17 3 98 4 32 7 56 9 24 11 23 13 19 14 18

Test Case 2

INPUT (STDIN)

15
0 1 2 0 1 1 1 0 2 0 2 2 0 1 2

EXPECTED OUTPUT

Link list data:0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
Link list data after fold:0 2 1 1 2 0 0 2 1 2 1 0 1 2 0

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

void print(struct node *head)

Test Case 3

KEYWORD

struct node *next;

Test Case 4

KEYWORD

void create(struct node **head,int data)

Complexity Test Cases

Test Case 1

Test Case 2

Test Case 3

CYCLOMATIC COMPLEXITY

2

TOKEN COUNT

510

NLOC

105

LinkedList8.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 void create(struct node **head, int data) {
10     struct node *new_node = (struct node*)malloc(sizeof(struct node));
11     new_node->data = data;
12     new_node->next = NULL;
13     if (*head == NULL) {
14         *head = new_node;
15         return;
16     }
17     struct node *temp = *head;
18     while (temp->next != NULL) {
19         temp = temp->next;
20     }
21     temp->next = new_node;
22 }
23
24 void print(struct node *head) {
25     printf("Link list data:");
```

```
26     struct node *temp = head;
27     while (temp != NULL) {
28         printf("%d ", temp->data);
29         temp = temp->next;
30     }
31     printf("\n");
32 }
33
34 void fold(struct node **head) {
35
36     int len = 0;
37     struct node *temp = *head;
38     while (temp != NULL) {
```

main | CH.SC.U4CSE25225 | LinkedList8.c

Ln 76, Col 1 Spaces: 4 UTF-8 CRLF C

```
40         temp = temp->next;
41     }
42     if (len <= 1) return;
```

```
43
44     int half = len / 2;
45     struct node *arr[len];
46     temp = *head;
47     for (int i = 0; i < len; i++) {
48         arr[i] = temp;
49         temp = temp->next;
50     }
51
52     for (int i = 0; i < half; i++) {
53         int t = arr[i]->data;
54         arr[i]->data = arr[len - 1 - i]->data;
55         arr[len - 1 - i]->data = t;
56     }
57 }
58
59 int main() {
```

```
60  int n;
61  if (scanf("%d", &n) != 1) return 0;
62
63  struct node *head = NULL;
64  for (int i = 0; i < n; i++) {
65      int val;
66      scanf("%d", &val);
67      create(&head, val);
68  }
69
70  print(head);
71  fold(&head);
72  printf("Link list data after fold:");
73  print(head);
74  return 0;
75}
76
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
16
```

```
1 3 4 7 9 11 13 14 18 19 23 24 56 32
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList8.exe
```

```
Link list data:1 3 4 7 9 11 13 14 18 19 23 24 56 32 32 32
```

```
Link list data after fold:Link list data:32 32 32 56 24 23 19 18 14 13 11 9 7 4 3 1
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
15
```

```
98 17 0 1 2 0 1 1 1 0 2 0 2 2 0 1 2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList8.exe
```

```
Link list data:98 17 0 1 2 0 1 1 1 0 2 0 2 2 0
```

```
Link list data after fold:Link list data:0 2 2 0 2 0 1 1 1 0 2 1 0 17 98
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```


DS	Session	Linked List	Question Information	Level 1	Challenge 39
----	---------	-------------	----------------------	---------	--------------

Question description

sanam's Dream came true after he got an Appointment order from Google.Simon's family was very happy of his achievement.

The company mentioned Basic Salary, DA, HRA with some other benefits.

But not highlighted the Gross salary in the order.

sanam's father wanted to know the Gross salary of his son.

sanam try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, Split a circular linked list in two halves. you have to split the circular Linked List with the same size of Divisions.Maybe if circular Linked List is odd, you have to change the number of node, it is even .

Can you help sanam?

Function Description

First count the number of node in Circular Linked List.

Second, you have to make the List even.

Third, you need to make the List half. the front is the same size like the rear. Finally, I have to make two circular Linked List.

Constraints

0<n<10

Input Format:

The First line represents the number of input in the circular linked list elements

Output Format:

First Line indicates the complete linked list

second line indicates the odd list

third line indicates the even list

Refer sample test cases.

Logical Test Cases

Test Case 1

INPUT (STDIN)

6

EXPECTED OUTPUT

Complete linked_list:
[h]==>1==>2==>3==>4==>5==>6==>[h]
Odd:
[h]==>1==>3==>5==>[h]
Even:
[h]==>2==>4==>6==>[h]

Test Case 2

INPUT (STDIN)

15

EXPECTED OUTPUT

Complete linked_list:
[h]==>1==>2==>3==>4==>5==>6==>7==>8==>9==>10==>11==>12==>13==>14==>15==>[h]
Odd:
[h]==>1==>3==>5==>7==>9==>11==>13==>15==>[h]
Even:
[h]==>2==>4==>6==>8==>10==>12==>14==>[h]

Mandatory Test Cases

Test Case 1

KEYWORD

void insert(int data)

Test Case 2

KEYWORD

struct n *next;

Test Case 3

KEYWORD

void display(struct n *h)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

620

Test Case 3

NLOC

115

LinkedList9.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *head = NULL;
10
11 void insert(int data) {
12     struct node *new_node = (struct node*)malloc(sizeof(struct node));
13     new_node->data = data;
14     new_node->next = NULL;
15     if (head == NULL) {
16         head = new_node;
17         head->next = head;
18         return;
19     }
20     struct node *temp = head;
21     while (temp->next != head) {
22         temp = temp->next;
23     }
24     temp->next = new_node;
25     new_node->next = head;
```

```

26}
27
28void display(struct node *h) {
29    if (h == NULL) {
30        printf("[h]=>\n");
31        return;
32    }
33    struct node *temp = h;
34    printf("[h]=>");
35    do {
36        printf("%d=>", temp->data);
37        temp = temp->next;
38    } while (temp != h);
39    printf("[h]\n");
40}
41
42void splitAlt(struct node **odd, struct node **even) {
43    struct node *temp = head;
44    int pos = 1;
45    do {

```

```
46     struct node *next = temp->next;
47     temp->next = NULL;
48     if (pos % 2 == 1) {
49         temp->next = *odd;
50         *odd = temp;
51     } else {
52         temp->next = *even;
53         *even = temp;
```

main | CH.SC.U4CSE25225 | LinkedList9.c

Ln 105, Col 1 Spaces: 4 UTF-8 CRLF C

```
54     }
55     temp = next;
56     pos++;
57 } while (temp != head);
58}
59
60 struct node *rev(struct node *h) {
61     struct node *p = NULL;
62     struct node *c = h;
63     while (c != NULL) {
64         struct node *n = c->next;
65         c->next = p;
```

```

66     p = c;
67     c = n;
68 }
69 return p;
70}
71
72void closeCircle(struct node *h) {
73     if (h == NULL) return;
74     struct node *t = h;
75     while (t->next != NULL) {
76         t = t->next;
77     }
78     t->next = h;
79}
80
81int main() {
82     int n;
83     if (scanf("%d", &n) != 1) return 0;
84
85     for (int i = 1; i <= n; i++) {

```

```
86     insert(i);
87 }
88
89 printf("Complete linked_list:\n");
90 display(head);
91
92 struct node *odd = NULL, *even = NULL;
93 splitAlt(&odd, &even);
94 odd = rev(odd);
95 even = rev(even);
96 closeCircle(odd);
97 closeCircle(even);
98
99 printf("Odd:\n");
100 display(odd);
101
102 printf("Even:\n");
103 display(even);
104 return 0;
105 }
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
6
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList9.exe
```

```
Complete linked_list:
```

```
[h]=>1=>2=>3=>4=>5=>6=>[h]
```

```
Odd:
```

```
[h]=>1=>3=>5=>[h]
```

```
Even:
```

```
[h]=>2=>4=>6=>[h]
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
15
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList9.exe
```

```
Complete linked_list:
```

```
[h]=>1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=>[h]
```

```
Odd:
```

```
[h]=>1=>3=>5=>7=>9=>11=>13=>15=>[h]
```

```
Even:
```

```
[h]=>2=>4=>6=>8=>10=>12=>14=>[h]
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```


Question description

Dr.Jegan is faculty, who handling data structure course for software engineering department second year students.

one day this faculty was handling very interesting topic in data structure such that Linked List, he has given the following explanation for Linked list concept.

"Linked List is a sequence of links which contains items. Each link contains a connection to another link. Linked list is the second most-used data structure after array. Following are the important terms to understand the concept of Linked List.

Link – Each link of a linked list can store a data called an element.

Next – Each link of a linked list contains a link to the next link called Next.

LinkedList – A Linked List contains the connection link to the first link called First."

During this lecture time, last bench students was asking surprise test for Linked list concept.

So the faculty decided to conduct test on the topic of Linked List.

the question was given to last bench students that is,

The nodes are deleted D times from the beginning of the given linked list.

For example if the given Linked List is 5->10->15->20->25 and remove 2 nodes,

then the Linked List becomes 15->20->25.

Constraint :

$1 < N < 1000$

$1 < P < N-1$

INPUT Format

First line contains the number of datas- N.

Second line contains N integers(the given linked list).

Third line contains no. of nodes to be deleted.

OUTPUT Format

Single line represents the final linked list after deletion.

▽ Logical Test Cases

Test Case 1

INPUT (STDIN)

4
9 77 12 6
2

EXPECTED OUTPUT

Linked List:->12->6

Test Case 2

INPUT (STDIN)

6
45 21 91 77 12 61
4

EXPECTED OUTPUT

Linked List:->12->61

▽ Mandatory Test Cases

Test Case 1

KEYWORD

```
struct node
```

Test Case 4

KEYWORD

```
for(i=0;i<n;i++)
```

Test Case 2

KEYWORD

```
node *next;
```

Test Case 3

KEYWORD

```
void create()
```

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

230

Test Case 3

NLOC

50

LinkedList10.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct node {
5     int data;
6     struct node *next;
7 };
8
9 struct node *head = NULL;
10
11 void create(int val) {
12     struct node *temp = (struct node *)malloc(sizeof(struct node));
13     temp->data = val;
14     temp->next = NULL;
15     if (head == NULL) {
16         head = temp;
17     } else {
18         struct node *p = head;
19         while (p->next != NULL) {
20             p = p->next;
21         }
22         p->next = temp;
23     }
24 }
25
```

```
26int main() {
27    int n, val, k, i;
28    if (scanf("%d", &n) != 1) return 0;
29
30    for (i = 0; i < n; i++) {
31        scanf("%d", &val);
32        create(val);
33    }
34
35    scanf("%d", &k);
36
37    struct node *p = head;
38    for (i = 0; i < k && p != NULL; i++) {
```

```
39     p = p->next;
40 }
41
42 printf("Linked List:");
43 while (p != NULL) {
44     printf("->%d", p->data);
45     p = p->next;
46 }
47 printf("\n");
48
49 return 0;
50}
51
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
4
```

```
9 77 12 6
```

```
2
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList10.exe
```

```
Linked List:->12->6
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
6
```

```
45 21 91 77 12 61
```

```
4
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>LinkedList10.exe
```

```
Linked List:->12->61
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```